

Tezpur University
Spring Semester Mid Examination - 2025

Course Code: CO503

Course Title: Fuzzy Logic and Neural Networks

Full marks = 30

Time = 1½ hours

(The figures in the right-hand margin indicate full marks for the questions)

1. State true or false: 1×5=5
 - (a) Set convexity is a cutworthy property.
 - (b) The expression: $[a, b] - [d, e] = [a-d, b-e]$ gives the rule for subtracting intervals.
 - (c) An AND gate can be implemented using a neural network having a single node.
 - (d) A neural network can perform feature extraction.
 - (e) All nodes in the same layer of a neural network may not use the same activation function.
2. Provide examples for each of the following: 1×5=5
 - (a) Fuzzy number representing approximately 5.
 - (b) Graphical representation of a large number defined on the universal set $[0, 100]$.
 - (c) Two crisp sets associated with a fuzzy set.
 - (d) Activation function producing fractional output in the range $[0, 1]$.
 - (e) Architecture of a single-layer recurrent neural network.
3. (a) How do we model an artificial neuron? How does the neuron compute its output? 2
 (b) How do you aggregate two or more fuzzy sets into a single fuzzy set? 2
 (c) How do you decompose a fuzzy set into crisp sets? 2
4. Answer the following questions based on fuzzy sets A, B defined on $X = \{-1, 0, 1, 2, \dots, 10\}$:

$$A = \frac{0.2}{1} + \frac{1.0}{2} + \frac{0.6}{3} + \frac{0.1}{4} \quad B = \frac{0.3}{2} + \frac{1.0}{4} + \frac{0.2}{6}$$
 - (a) Provide the definition of a fuzzy complement other than the standard one. Apply this definition to compute the complement of fuzzy set A . 2
 - (b) Find $A \cap B$ and $A \cup B$ using the t-norm and the t-conorm defined below: 2

$$i(a, b) = ab \quad \text{and} \quad u(a, b) = a + b - ab \quad \text{where, } a, b \in [0, 1].$$
 - (c) Consider the function $f(x) = x^2 - 3x + 1$ mapping a real number to another real number i.e. $f: R \rightarrow R$. Find $f(A)$ using the extension principle. 2
 - (d) Compute $A + B$. 3
5. Consider the 2-layer feedforward neural network having 2, 3 and 2 nodes in the input layer, hidden layer and output layer respectively. The weight matrices of the two layers are given below:

$$w_{ij} = \begin{bmatrix} 2 & 1 & 4 \\ 5 & 2 & 1 \end{bmatrix} \quad v_{jk} = \begin{bmatrix} 2 & 2 \\ 3 & 3 \\ 3 & 5 \end{bmatrix}$$
 - (a) Draw the architecture of the network. Mark the links with the corresponding weights. 2
 - (b) Compute the binary output vector for the input vector $[1, -1]$. Use the activation function: 3

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$